Ramsey El-Naggar

Software Engineer

Contact

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Languages

- English: Native
- Spanish: B1 (DELE, Sep 2021)

Skills

Main languages (and frameworks / versions) in order of proficiency:

1. Python (Django)
2. PHP
3. Java
4. SQL (MySQL, PostgreSQL)
5. HTML/CSS (Bootstrap)
6. Ruby (Ruby on Rails)
7. JavaScript

Additional languages/tools:

- Git, Bash, Docker, AWS
- Apache HTTP Server
- MATLAB, C, Haskell

Coursework

Compiling Techniques (100%)
Implemented (in Java) sections of a compiler for a subset of C, including: lexing, parsing, semantic analysis and Java bytecode generation.

System Design Project (88%)
Led a group of four to design and build a football-playing robot, placing second in the end-of-year competition, scoring six goals, and winning the **prize for the best technical presentation in the year**.

Education

Master of Informatics (MInf, CEng, CITP, CSci)

The University of Edinburgh | 2013 - 2018 (5 years)

- Degree classification: **First Class Honours (*1st*)**
- Dissertation (*82%*): A biorobotic investigation of the visual mechanisms controlling dung beetle celestial menotaxis.
- **Built a dung beetle robot**, programmed the robot with a neural model of navigational behaviour, and received a grant to travel to South Africa and test the robot against real dung beetles.
- Software engineering courses: Data and Analysis (*99%*), Elements of Programming Languages (*98%*), Object-Oriented Programming (*97%*), Computer Security (*92%*), Computer Communication and Networking (*90%*), Computer Systems (*90%*), Algorithms, Data Structures and Learning (*86%*), Software Engineering (*83%*).

Experience

Computer Science and Programming Tutor

Self-employed | Jan 2019 - Present (4 years, 1 month)

- Currently the top 'Computing University' tutor in the UK on the leading tuition platform MyTutor, with **over 600 5-star ratings and reviews**.
- Taught topics including web development (HTML/CSS, Django, PHP, Ruby on Rails), object-oriented programming and software engineering (Python, Java), databases (MySQL), algorithms and data structures.

Full Stack Developer

N-able | Jun 2015 - Aug 2015 (3 months)

- Upgraded and improved a large existing codebase.
- Front-end: upgraded plain JavaScript to Ext JS (a JS framework).
- Back-end: uncoupled the domain logic from the database layer and **wrote a custom object-relational mapping** (ORM) in PHP and SQL (MySQL), thus upgrading the procedural PHP to object-oriented PHP.
- Testing: introduced Jasmine and PHPUnit to the codebase in order to use test-driven development (TDD) throughout the upgrade process.

Projects

<rmsy.me>: A personal website

rmsy.me/github

To showcase my front- and back-end development skills.

- Built a **reusable back-end PHP framework** from scratch.
- Designed a **responsive** front-end with Bootstrap.
- Used **Docker** to organise the application into standalone containers.
- **Automated the entire DevOps pipeline** via Bash scripts for each stage.
- Manually configured the AWS EC2 instance and the Apache Server.

Extras

- British Computer Society (BCS) Prize for **Professional Issues** (*99%*).
- University of Edinburgh tutor for six different Informatics courses and three-times nominated for the **Best Student Who Tutors Award**.
- Edinburgh Award for volunteering for two years to lead peer-assisted learning sessions; **recruited, trained and managed 12 more leaders**.
- Musical director at the Edinburgh University A Cappella Society, placing in the **semi-finals of the national competition** Voice Festival UK.